

Computer Organization and Architecture		
Lab #:	3	
Lab Title:	Calculate Numbers	
Submitted by:		
Names		Registration #
Sumaira Safeer		FA22-BCE-019

Rubrics name & number					Marks	
					In-Lab	Post-Lab
Engineering Knowledge	R2: Use of Engineering Knowledge and follow Experiment Procedures: Ability to follow experimental procedures, control variables, and record procedural steps on lab report.					
Problem Analysis	R6: Experimental Data Analysis: Ability to interpret findings, compare them to values in the literature, identify weaknesses and limitations.					
Design	R8: Best Coding Standards: Ability to follow the coding standards and programming practices.					
Modern Tools Usage	R9: Understand Tools: Ability to describe and explain the principles behind and applicability of engineering tools.					
Individual and Teamwork	R12: Individual Work Contributions: Ability to carry out individual responsibilities.					
	R13: Management of Team Work: Ability to appreciate, understand and work with multidisciplinary team members.					
Rubrics #	R2	R6	R8	R9	R12	R13
In Lab						
Post- Lab						

Task No: 01

Write a program that inputs marks of four quizzes and four assignments, each out of 10, and the calculates total marks out of 25. Total quiz marks 15/25 and Assignment marks = 10/25

Code:

```
.data
Quiz_prompt: .asciiz "Enter marks for Quiz: "
Assign_prompt: .asciiz "Enter marks for Assignment: "
Quiz_total: .asciiz "Total Quiz Marks (out of 15): "
Assign_total: .asciiz "Total Assignment Marks (out of 10): "
Overall_Result: .asciiz "Overall Total Marks (out of 25): "
newline: .asciiz "\n"
.text
.globl main

main:
    # Initialize total marks for quizzes and assignments
    li $t0, 0 # total_quiz_marks
    li $t1, 0 # total_assignment_marks

    # Input marks for quizzes
    li $t2, 4 # loop counter for quizzes
    la $a0, Quiz_prompt
quiz_loop:
    li $v0, 4
    syscall
    li $v0, 5
    syscall
```

```

add $t0, $t0, $v0
sub $t2, $t2, 1
bnez $t2, quiz_loop

# Input marks for assignments
li $t3, 4 # loop counter for assignments
la $a0, Assign_prompt
assign_loop:
    li $v0, 4
    syscall
    li $v0, 5
    syscall
    add $t1, $t1, $v0
    sub $t3, $t3, 1
    bnez $t3, assign_loop

# Calculate total marks for quizzes (out of 15)
li $t4, 15
mul $t0, $t0, $t4
li $t5, 40
div $t0, $t0, $t5

# Calculate total marks for assignments (out of 10)
li $t6, 10
mul $t1, $t1, $t6
div $t1, $t1, $t5

# Calculate overall total marks (out of 25)
add $t7, $t0, $t1

```

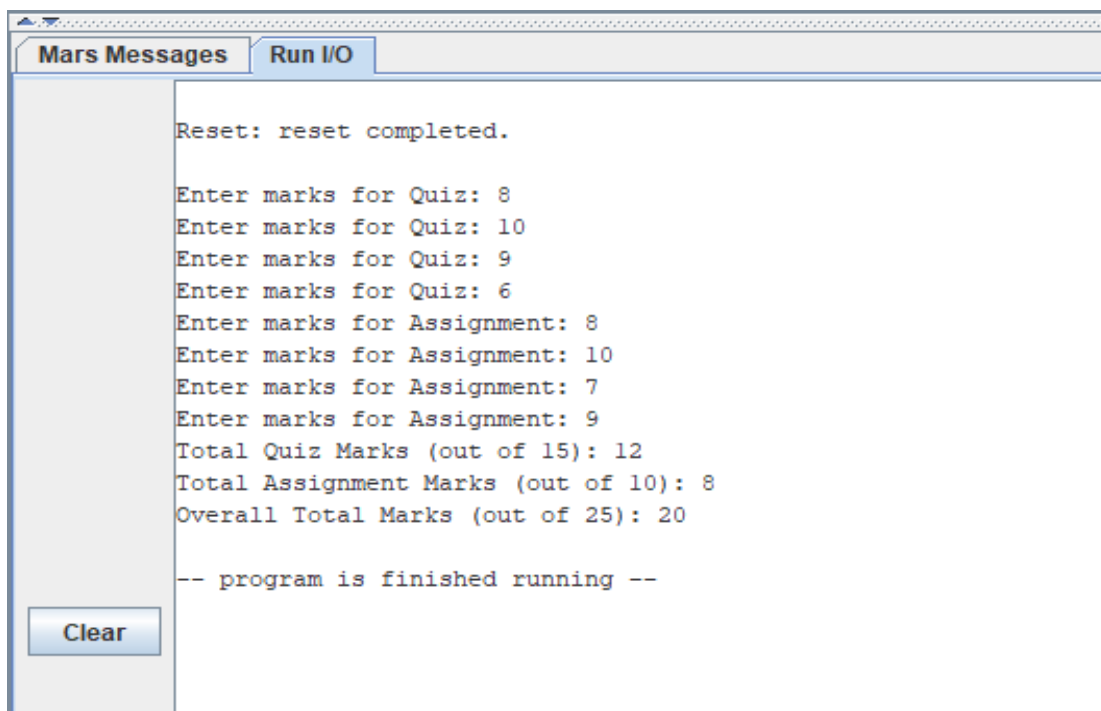
```
# Print total quiz marks
la $a0, Quiz_total
li $v0, 4
syscall
move $a0, $t0
li $v0, 1
syscall
la $a0, newline # newline for move to next line
li $v0, 4
syscall
```

```
# Print total assignment marks
la $a0, Assign_total
li $v0, 4
syscall
move $a0, $t1
li $v0, 1
syscall
la $a0, newline
li $v0, 4
syscall
```

```
# Print overall total marks
la $a0, Overall_Result
li $v0, 4
syscall
move $a0, $t7
li $v0, 1
```

```
syscall  
la $a0, newline  
li $v0, 4  
syscall  
  
# Exit program  
li $v0, 10  
syscall
```

Output:



Run speed at max (no int)																																																											
Edit Execute																																																											
Text Segment <table> <tr> <th>Bkpt</th><th>Address</th><th>Code</th><th>Basic</th><th>Source</th></tr> <tr> <td>☐</td><td>4194304</td><td>0x24080000</td><td>addiu \$8,\$0,0x00000000</td><td>14: li \$t0, 0 # total_quiz_marks</td></tr> <tr> <td>☐</td><td>4194308</td><td>0x24090000</td><td>addiu \$9,\$0,0x00000000</td><td>15: li \$t1, 0 # total_assignment_marks</td></tr> <tr> <td>☐</td><td>4194312</td><td>0x240a0004</td><td>addiu \$10,\$0,0x0000...</td><td>18: li \$t2, 4 # loop counter for quizzes</td></tr> <tr> <td>☐</td><td>4194316</td><td>0x3c011001</td><td>lui \$1,0x00001001</td><td>19: la \$a0, Quiz_prompt</td></tr> <tr> <td>☐</td><td>4194320</td><td>0x34240000</td><td>ori \$4,\$1,0x00000000</td><td></td></tr> <tr> <td>☐</td><td>4194324</td><td>0x24020004</td><td>addiu \$2,\$0,0x00000004</td><td>21: li \$v0, 4</td></tr> <tr> <td>☐</td><td>4194328</td><td>0x0000000c</td><td>syscall</td><td>22: syscall</td></tr> <tr> <td>☐</td><td>4194332</td><td>0x24020005</td><td>addiu \$2,\$0,0x00000005</td><td>23: li \$v0, 5</td></tr> <tr> <td>☐</td><td>4194336</td><td>0x0000000c</td><td>syscall</td><td>24: syscall</td></tr> <tr> <td>☐</td><td>4194340</td><td>0x01024020</td><td>add \$8,\$8,\$2</td><td>25: add \$t0, \$t0, \$v0</td></tr> </table>					Bkpt	Address	Code	Basic	Source	☐	4194304	0x24080000	addiu \$8,\$0,0x00000000	14: li \$t0, 0 # total_quiz_marks	☐	4194308	0x24090000	addiu \$9,\$0,0x00000000	15: li \$t1, 0 # total_assignment_marks	☐	4194312	0x240a0004	addiu \$10,\$0,0x0000...	18: li \$t2, 4 # loop counter for quizzes	☐	4194316	0x3c011001	lui \$1,0x00001001	19: la \$a0, Quiz_prompt	☐	4194320	0x34240000	ori \$4,\$1,0x00000000		☐	4194324	0x24020004	addiu \$2,\$0,0x00000004	21: li \$v0, 4	☐	4194328	0x0000000c	syscall	22: syscall	☐	4194332	0x24020005	addiu \$2,\$0,0x00000005	23: li \$v0, 5	☐	4194336	0x0000000c	syscall	24: syscall	☐	4194340	0x01024020	add \$8,\$8,\$2	25: add \$t0, \$t0, \$v0
Bkpt	Address	Code	Basic	Source																																																							
☐	4194304	0x24080000	addiu \$8,\$0,0x00000000	14: li \$t0, 0 # total_quiz_marks																																																							
☐	4194308	0x24090000	addiu \$9,\$0,0x00000000	15: li \$t1, 0 # total_assignment_marks																																																							
☐	4194312	0x240a0004	addiu \$10,\$0,0x0000...	18: li \$t2, 4 # loop counter for quizzes																																																							
☐	4194316	0x3c011001	lui \$1,0x00001001	19: la \$a0, Quiz_prompt																																																							
☐	4194320	0x34240000	ori \$4,\$1,0x00000000																																																								
☐	4194324	0x24020004	addiu \$2,\$0,0x00000004	21: li \$v0, 4																																																							
☐	4194328	0x0000000c	syscall	22: syscall																																																							
☐	4194332	0x24020005	addiu \$2,\$0,0x00000005	23: li \$v0, 5																																																							
☐	4194336	0x0000000c	syscall	24: syscall																																																							
☐	4194340	0x01024020	add \$8,\$8,\$2	25: add \$t0, \$t0, \$v0																																																							

Registers	Coproc 1	Coproc 0
Name	Number	Value
\$zero	0	0x00000000
\$at	1	0x10010000
\$v0	2	0x00000000
\$v1	3	0x00000000
\$a0	4	0x00000000
\$a1	5	0x00000000
\$a2	6	0x00000000
\$a3	7	0x00000000
\$t0	8	0x00000000
\$t1	9	0x00000000
\$t2	10	0x00000004
\$t3	11	0x00000000
\$t4	12	0x00000000
\$t5	13	0x00000000
\$t6	14	0x00000000
\$t7	15	0x00000000
\$s0	16	0x00000000

Task No: 02

Write a program that inputs three 1-digit numbers and combines them to form a 3 digit number. For example input 4, 8 ,7, output=487.

Code:

```
.data
prompt1: .asciiz "Enter the first digit: "
prompt2: .asciiz "Enter the second digit: "
prompt3: .asciiz "Enter the third digit: "
Result: .asciiz "The combined 3-digit number WILL BE: "
```

```

newline: .asciiz "\n"

.text
.globl main

main:
    # Input first digit
    la $a0, prompt1
    li $v0, 4
    syscall
    li $v0, 5
    syscall
    move $t0, $v0 # Store first digit in $t0

    # Input second digit
    la $a0, prompt2
    li $v0, 4
    syscall
    li $v0, 5
    syscall
    move $t1, $v0 # Store second digit in $t1

    # Input third digit
    la $a0, prompt3
    li $v0, 4
    syscall
    li $v0, 5
    syscall
    move $t2, $v0 # Store third digit in $t2

    # Combine digits to form a 3-digit number
    mul $t0, $t0, 100 # Multiply first digit by 100

```

```

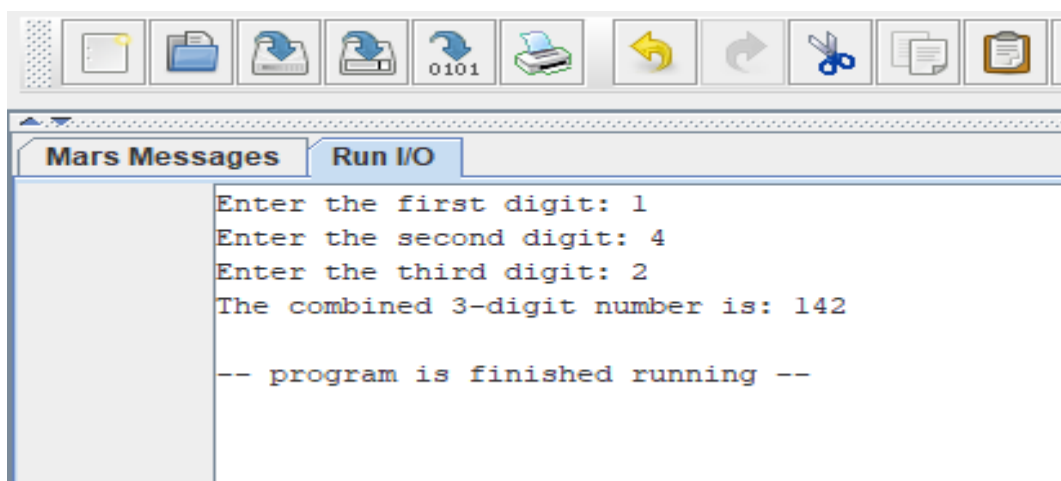
mul $t1, $t1, 10 # Multiply second digit by 10
add $t3, $t0, $t1 # Add first and second digits
add $t3, $t3, $t2 # Add third digit to get the final number

# Print the result
la $a0, Result
li $v0, 4
syscall
move $a0, $t3
li $v0, 1
syscall
la $a0, newline
li $v0, 4
syscall

# Exit program
li $v0, 10
syscall

```

Output:



```

Enter the first digit: 1
Enter the second digit: 4
Enter the third digit: 2
The combined 3-digit number is: 142

-- program is finished running --

```


					Run spe
Edit Execute					
Text Segment					
Bkpt	Address	Code	Basic	Source	
☐	4194420	0x24020004	addiu \$2,\$0,0x00000004	44:	li \$v0, 4
☐	4194424	0x0000000c	syscall	45:	syscall
☐	4194428	0x000b2021	addu \$4,\$0,\$11	46:	move \$a0, \$t3
☐	4194432	0x24020001	addiu \$2,\$0,0x00000001	47:	li \$v0, 1
☐	4194436	0x0000000c	syscall	48:	syscall
☐	4194440	0x3c011001	lui \$1,0x00001001	49:	la \$a0, newline
☐	4194444	0x3424006a	ori \$4,\$1,0x0000006a		
☐	4194448	0x24020004	addiu \$2,\$0,0x00000004	50:	li \$v0, 4
☐	4194452	0x0000000c	syscall	51:	syscall
☐	4194456	0x2402000a	addiu \$2,\$0,0x0000000a	54:	li \$v0, 10

Registers		Coproc 1	Coproc 0
Name	Number	Value	
\$zero	0	0x00000000	
\$at	1	0x10010000	
\$v0	2	0x00000004	
\$v1	3	0x00000000	
\$a0	4	0x1001006a	
\$a1	5	0x00000000	
\$a2	6	0x00000000	
\$a3	7	0x00000000	
\$t0	8	0x00000064	
\$t1	9	0x00000028	
\$t2	10	0x00000002	